

ANALYSIS QUALIFYING EXAM MAY 2025

Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

DO SIX OF THE EIGHT PROBLEMS.

- (1) For any two of the following, either give an example or explain why no example exists.
 - (i) A Lebesgue measurable set $E \subseteq [0, 1]$ that is closed, has positive measure, but contains no non-trivial interval.
 - (ii) A Lebesgue measurable set $E \subseteq [0, 1]$ of positive measure such that $E - E = \{x - y : x, y \in E\}$ contains no non-trivial interval.
 - (iii) A measure space $(X, \mathcal{M})$ and a set $E \subseteq X \times X$ such that $[E]_x, [E]^y \in \mathcal{M}$ for all $x, y \in X$, but $E \notin \mathcal{M} \otimes \mathcal{M}$.
- (2) Suppose X, Y are metric spaces. Show, if $f : X \rightarrow Y$ is continuous, then f is Borel measurable.
- (3) Let $I = (-1, 1)$ and let m^* denote Lebesgue outer measure. Show, if $A \subseteq I$ and $m^*(A) + m^*(I \setminus A) = 2$, then A is Lebesgue measurable.
- (4) State the Radon-Nikodym Theorem and give an example that shows the σ -finite hypothesis is needed.
- (5) Suppose $\mathcal{X}$ is a Banach space and $\mathcal{M}$ and $\mathcal{N}$ are (closed) subspaces of $\mathcal{X}$. Show, if each $x \in \mathcal{X}$ has a unique representation as $x = m + n$ for some $m \in \mathcal{M}$ and $n \in \mathcal{N}$, then the mapping $\mathcal{X} \rightarrow \mathcal{M}$ sending $x = m + n$ to m is linear and bounded.
- (6) Prove, if $\mathcal{X}$ is a Banach space and $\mathcal{X}^*$ is separable, then $\mathcal{X}$ is separable. Is ℓ^1 isomorphic, as a Banach space, to $(\ell^\infty)^*$?

(7) Short answer. Do two.

- (i) Suppose $h : \mathbb{R} \rightarrow \mathbb{C}$ is a measurable function. If $hf \in L^1(\mathbb{R})$ for every $f \in L^4(\mathbb{R})$, what can be said about h ? Explain your answer.
- (ii) Does there exist a (bounded) linear functional $L : \ell^\infty(\mathbb{N}) \rightarrow \mathbb{C}$ such that $L(f) = \lim f(n)$ whenever $f \in \ell^\infty(\mathbb{N})$ and $(f(n))_n$ converges? Explain your answer.
- (iii) Does there exist an inner product on $\mathbb{R}^2$ that induces the norm $\|x\| = |x_1| + |x_2|$ for $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2$. Explain your answer.

(8) Do two providing short justifications for your answers. Here $\widehat{}$ denotes the Fourier transform of an L^1 function and $\mathcal{F}f$ denotes the Fourier transform of an $L^2(\mathbb{R})$ function f .

- (i) Define $f \in L^1(\mathbb{R})$ by $f(x) = e^{-x^4} \chi_{[1, \infty)}(x)$. Is $\widehat{f} \in L^1(\mathbb{R})$?
- (ii) Recall, if $f \in L^1(\mathbb{R})$, then $\|\widehat{f}\|_\infty \leq \|f\|_1$. When does equality hold?
- (iii) Does there exist $0 \neq f \in L^2(\mathbb{R})$ such that $\mathcal{F}f = e^{i\frac{\pi}{4}} f$?